P1 - Drought Severity Assessment of the Bundelkhand Region Using Multi-Temporal Vegetation Condition Index (VCI) Derived from MODIS Satellite Data (2015–2023)

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

Subject: Remote Sensing and GIS with Python

By

Anshuman Tripathi
Bundelkhand University
Institute Roll No.: 221381030012
Enrollment No.: 688677

Under the Supervision of
Miss. Alisha Sinha

India Space Academy
Department of Space Education, India Space Week
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1. Title

Drought Severity Assessment of the Bundelkhand Region Using Multi-Temporal Vegetation Condition Index (VCI) Derived from MODIS Satellite Data (2015–2023).

Project Code: P1 — Drought Severity Mapping using Vegetation Condition Index (VCI).

2. Objective

The main aim of this project is to look at how drought conditions have changed over the Bundelkhand region between 2015 and 2023, using satellite-based vegetation data. Bundelkhand is one of the most drought affected parts of central India, and farmers here depend heavily on rainfall for their crops. So the idea was to see if satellite images can actually tell us something useful about how bad the drought was in a particular year, and how it compares with other years.

To do this, MODIS NDVI data for nine years (2015 to 2023) was collected and processed to work out the Vegetation Condition Index (VCI) for every pixel in the study area. VCI basically compares how green the vegetation is in a given year against the driest and the greenest it has ever been over the whole time period, which gives a number between 0 and 100. Lower VCI values mean the vegetation is stressed, which usually points towards drought.

Along with calculating VCI and classifying the region into drought categories, the project also checks whether the VCI values match up with the actual rainfall recorded by CHIRPS for the same years. This helps in understanding whether vegetation stress is only driven by rainfall or if other factors are also involved.

3. Study Area

Bundelkhand is a semi-arid region located in central India, spread across the states of Uttar Pradesh and Madhya Pradesh. It is well known for its rocky, undulating terrain and for facing drought almost every few years because of unreliable and poorly distributed rainfall. For this study, the boundary of Bundelkhand was created by combining 13 districts using the FAO GAUL 2015 Level-2 administrative dataset inside Google Earth Engine.

Seven of these districts fall in Uttar Pradesh — Jhansi, Lalitpur, Jalaun, Hamirpur, Mahoba, Banda and Chitrakoot — while the remaining six are in Madhya Pradesh — Datia, Tikamgarh, Chhatarpur, Panna, Damoh and Sagar. These 13 district polygons were merged and dissolved into one single Area of Interest (AOI), which was then simplified slightly (100 m tolerance) to remove small topology errors that were causing projection problems later while clipping MODIS images.

The final AOI roughly extends from about 78.05°E to 81.56°E in longitude and from about 23.15°N to 26.45°N in latitude, covering a large stretch of central India. Figure 1 shows the district boundaries and the merged AOI used for the whole analysis.

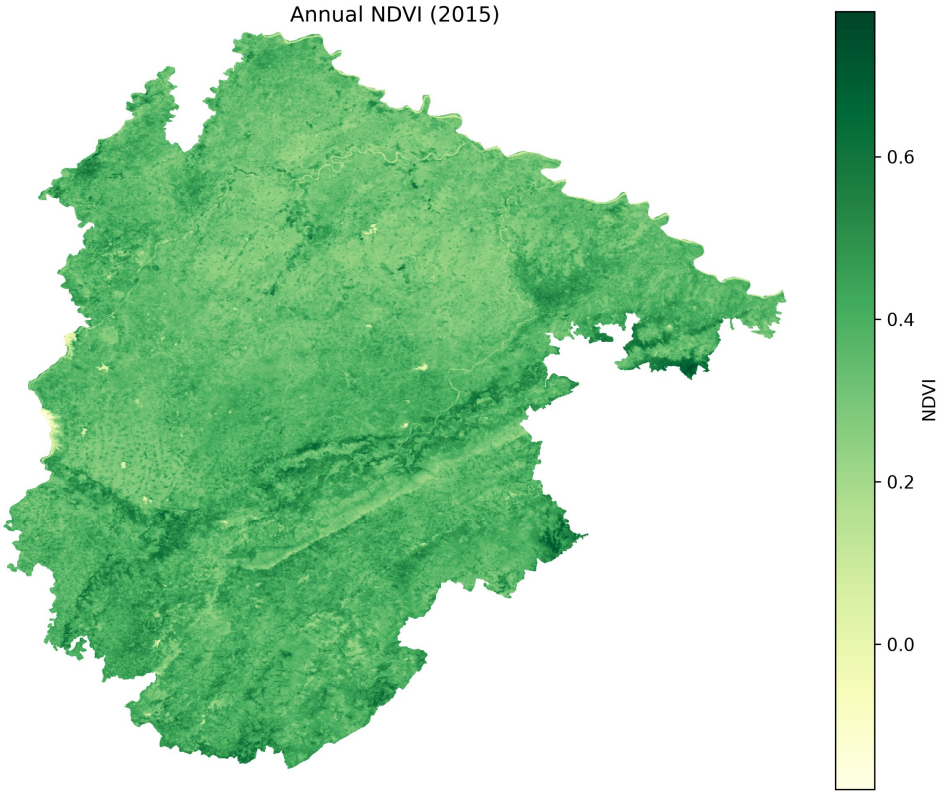

Figure 1: Bundelkhand Area of Interest (AOI) formed by merging 13 districts of Uttar Pradesh and Madhya Pradesh (FAO GAUL 2015, Level-2).

4. Data Used

The following datasets were used in this project:

- MODIS Terra Vegetation Indices, MOD13Q1 Version 6.1 — 250 m spatial resolution, 16-day NDVI composites, accessed through the Google Earth Engine Data Catalog (collection: MODIS/061/MOD13Q1). Data for the period 1 January 2015 to 31 December 2023 was used, which gave 207 individual scenes for the AOI.
- CHIRPS Daily Rainfall data — used as reference/validation data to compare against the calculated VCI values for the same nine years.
- FAO GAUL 2015 Level-2 administrative boundary dataset — used only to build the Bundelkhand AOI by selecting and merging the 13 districts mentioned above.

The MODIS NDVI band was first masked using the SummaryQA band (keeping only pixels marked as Good = 0 or Marginal = 1) and an additional fill-value mask was applied to remove the -3000 no-data value that MOD13Q1 sometimes carries. After masking, the NDVI values were scaled using the standard MODIS scale factor of 0.0001. All the processing after this point (VCI calculation, drought classification, correlation analysis) was carried out in Python using libraries such as NumPy, Pandas, Rasterio, GeoPandas, Matplotlib, SciPy and scikit-learn inside a Jupyter Notebook.

5. Methodology

The methodology for this project was split into two parts. The first part, done in Google Earth Engine (GEE), was used to prepare the AOI and to generate and export the yearly NDVI composites. The second part, done in Python, used those exported NDVI rasters to calculate VCI, classify drought severity, and check the relationship between VCI and rainfall.

5.1 Google Earth Engine (GEE) Based Workflow

- Loaded the FAO GAUL 2015 Level-2 FeatureCollection and filtered it for India, then selected the 13 required districts from Uttar Pradesh and Madhya Pradesh.
- Merged the district features and dissolved them into a single AOI geometry using the union() function, then simplified the geometry (100 m error margin) to remove small sliver artefacts left behind after the union.
- Loaded the MODIS MOD13Q1 collection, filtered it by the AOI and by date (2015–2023), and applied the SummaryQA and fill-value masks described above.
- Scaled the NDVI band by 0.0001 and generated one median composite for every year (9 composites total), reprojecting each composite to EPSG:4326 at 250 m before clipping it to the AOI. Reprojecting before clipping avoided a recurring CRS transform error that occurs when Earth Engine tries to clip a Sinusoidal-projection MODIS image directly with a WGS-84 polygon.
- Computed zonal statistics (mean, minimum, maximum, standard deviation) of NDVI for each year over the AOI and exported both the 9 annual NDVI GeoTIFFs and the statistics table (CSV) to Google Drive for use in the next stage.

5.2 Python Based Workflow

- Read and validated all nine annual NDVI GeoTIFFs (checked resolution, CRS, dimensions and bounding box to make sure all rasters lined up on the same grid).
- Stacked the nine rasters into a single 3-D NumPy array and computed the pixel-wise minimum (NDVImin) and pixel-wise maximum (NDVImax) NDVI values across the full 2015–2023 period.
- Calculated the Vegetation Condition Index for every pixel and every year using the standard formula:

$$VCI = [(NDVI - NDVImin) / (NDVImax - NDVImin)] \times 100$$

The resulting VCI values were clipped between 0 and 100 to remove any minor rounding errors. VCI values close to 0 indicate severe vegetation stress (drought-like condition), while values closer to 100 indicate the vegetation is close to its healthiest recorded state.

- Classified the VCI raster of each year into five drought severity classes, based on the widely used Kogan classification scheme: Extreme Drought (VCI 0–20), Severe Drought (20–40), Moderate Drought (40–60), Mild Drought (60–80) and Normal (VCI ≥ 80).
- Calculated the area (in km²) covered by each drought class for every year, and also converted the classified drought raster into vector polygons for mapping purposes.
- Loaded the CHIRPS annual rainfall rasters and computed the mean annual rainfall over the AOI for each of the nine years.
- Compared the mean annual VCI against mean annual rainfall using Pearson correlation, Spearman rank correlation and simple linear regression, to check if wetter years actually showed healthier vegetation (higher VCI) in this region.
- Generated publication-quality maps for NDVI, VCI and drought classification for each year, along with a combined trend-analysis dashboard and a VCI-vs-rainfall correlation plot.

6. Results

This section presents the year-wise NDVI, VCI and drought classification results for Bundelkhand, along with the comparison between VCI and CHIRPS rainfall.

6.1 NDVI Minimum and Maximum Composites

Figures 2 and 3 show the pixel-wise minimum and maximum NDVI recorded across the 2015–2023 period. These two composites form the base reference used to calculate VCI for every year — the driest condition a pixel has ever shown (NDVImin) and the greenest condition it has ever shown (NDVImax).

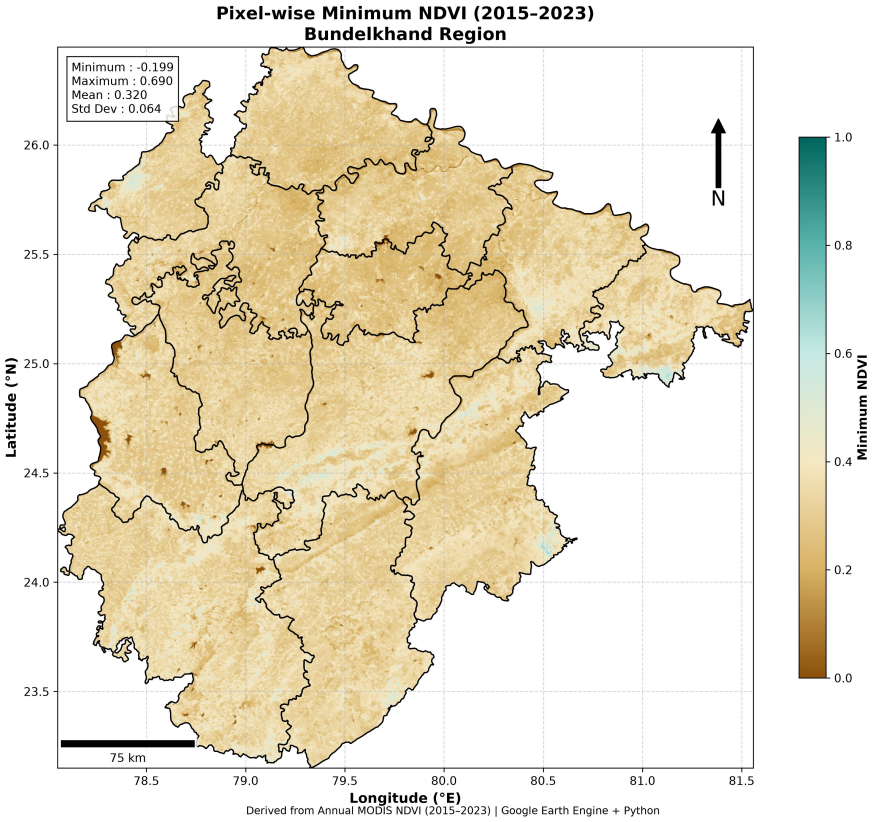

Figure 2: Pixel-wise minimum NDVI (2015–2023) across the Bundelkhand AOI.

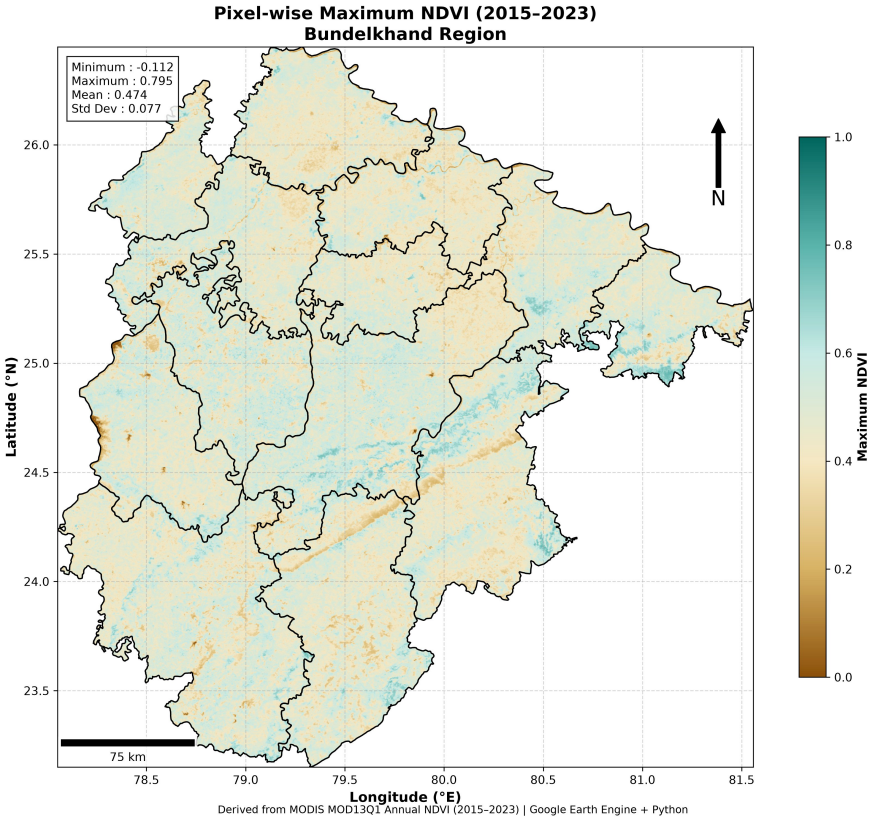

Figure 3: Pixel-wise maximum NDVI (2015–2023) across the Bundelkhand AOI.

6.2 Annual NDVI and VCI Trend

Table 1 lists the mean NDVI, mean VCI and mean annual rainfall for each year from 2015 to 2023. Looking at the numbers, 2016 and 2018 stand out as the worst years for vegetation health, with mean VCI values of about 18 and 19 respectively — both fall in the severe drought range. From 2020 onward, the region shows a clear recovery, with VCI climbing to above 70 in 2020, and finally reaching almost 77 in 2023, the highest of the whole nine-year record.

Year	Mean NDVI	Mean VCI	Mean Rainfall (mm)
2015	0.370	34.3	753
2016	0.346	18.2	1351
2017	0.392	48.2	855
2018	0.349	19.2	1030
2019	0.373	34.8	1362
2020	0.427	71.1	946
2021	0.408	57.2	994
2022	0.421	65.9	1194
2023	0.440	76.9	856

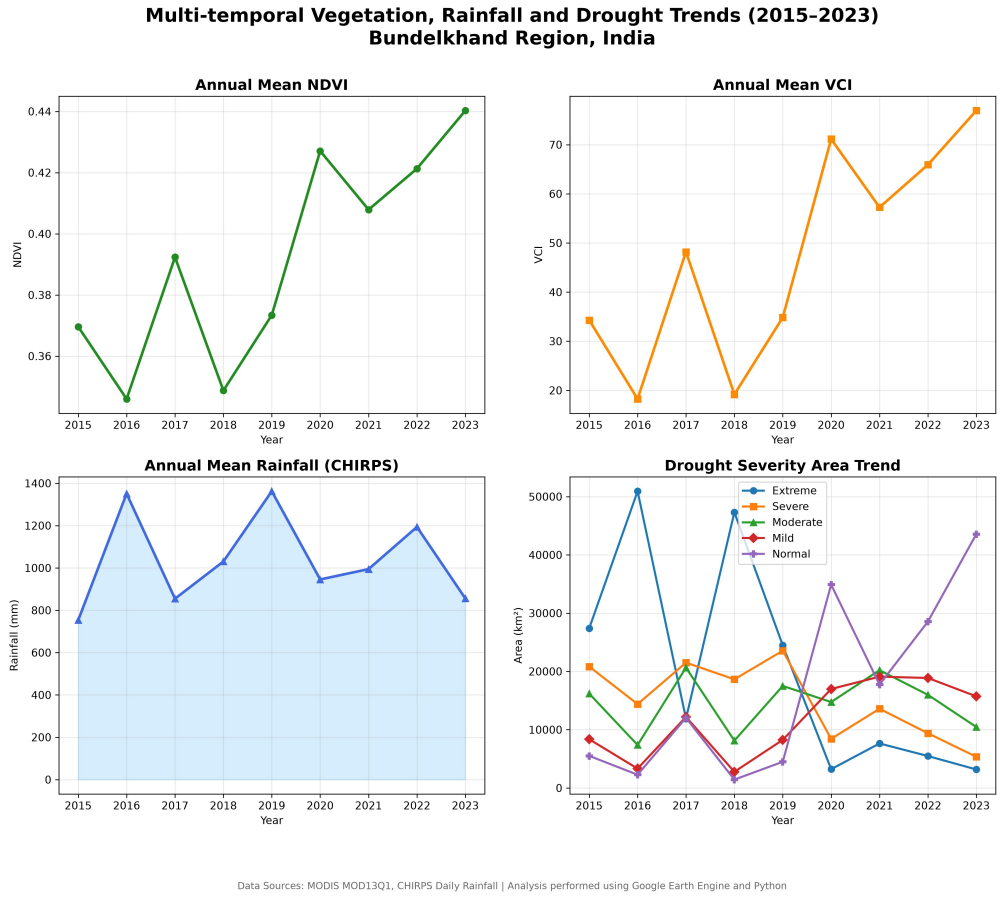

Figure 4: Multi-temporal trend dashboard showing NDVI, VCI, rainfall and drought class area for 2015–2023.

6.3 Drought Severity Classification

Once VCI was classified into five severity classes, the area under each class was calculated for every year (Table 2). The year 2016 had by far the largest extreme-drought area, close to 50,923 km², which is roughly half of the entire Bundelkhand region under extreme vegetation stress. 2018 was almost as bad, with about 47,312 km² under extreme drought. On the other hand, 2023 had only 3,172 km² under extreme drought and about 43,560 km² classified as Normal, making it the healthiest year in the record.

Year	Extreme (km²)	Severe (km²)	Moderate (km²)	Mild (km²)	Normal (km²)
2015	27,392	20,833	16,191	8,356	5,501
2016	50,923	14,389	7,367	3,319	2,273
2017	11,892	21,509	20,648	12,193	12,031
2018	47,312	18,640	8,120	2,768	1,432
2019	24,476	23,548	17,511	8,247	4,487
2020	3,219	8,427	14,742	16,982	34,903

Year	Extreme (km²)	Severe (km²)	Moderate (km²)	Mild (km²)	Normal (km²)
2021	7,633	13,591	20,210	19,103	17,735
2022	5,476	9,385	15,984	18,872	28,555
2023	3,172	5,352	10,466	15,722	43,560

Figures 5 and 6 compare the VCI map and drought severity map of 2016 (the worst year) with the same maps for 2023 (the best year). The contrast between the two years is quite visible — most of the region turns dark red (extreme/severe drought) in 2016, while in 2023 the map is dominated by green tones (normal/mild).

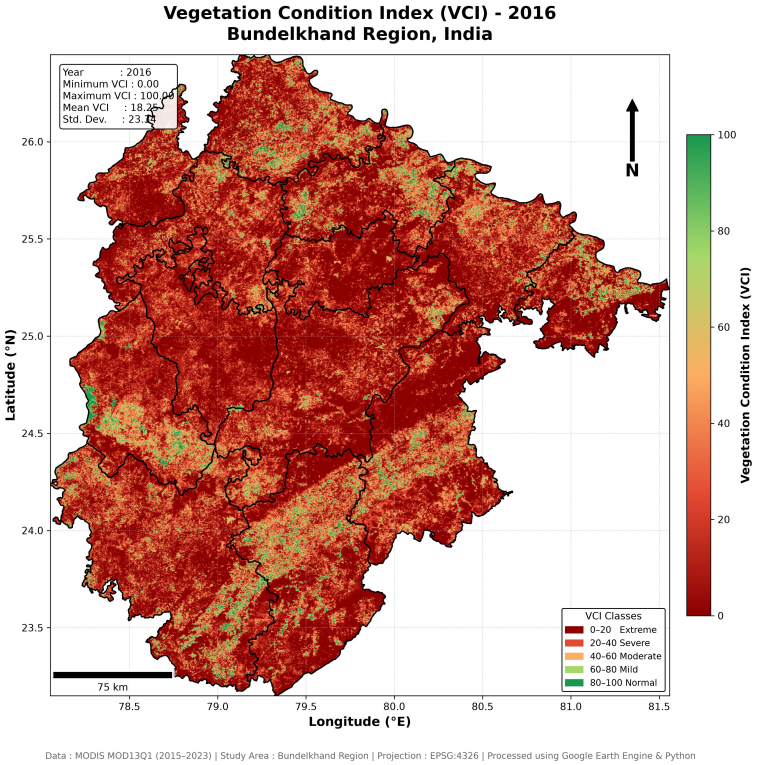

Figure 5a: VCI map for 2016, the year with the most widespread drought stress.

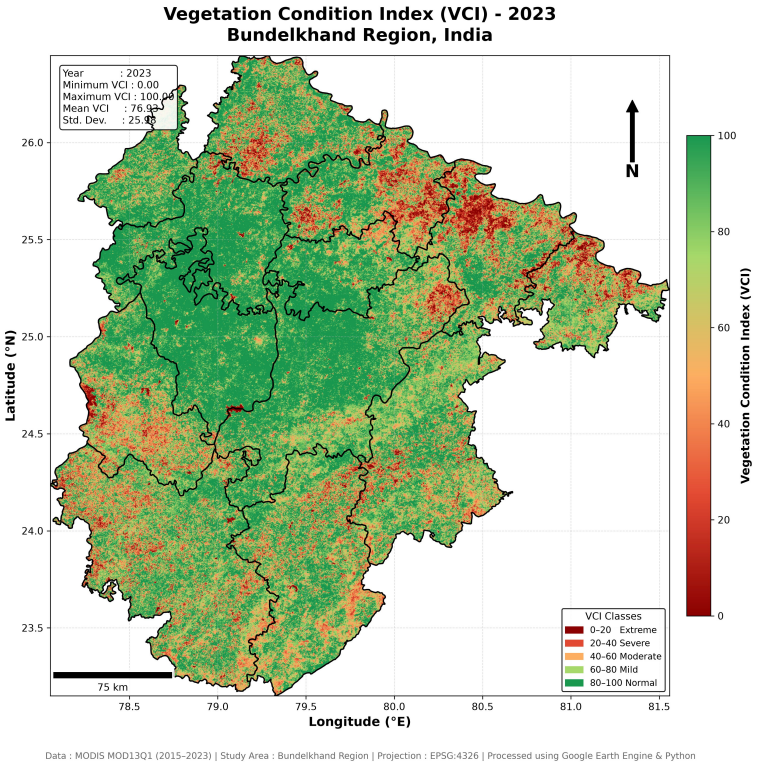

Figure 5b: VCI map for 2023, the year with the healthiest vegetation condition.

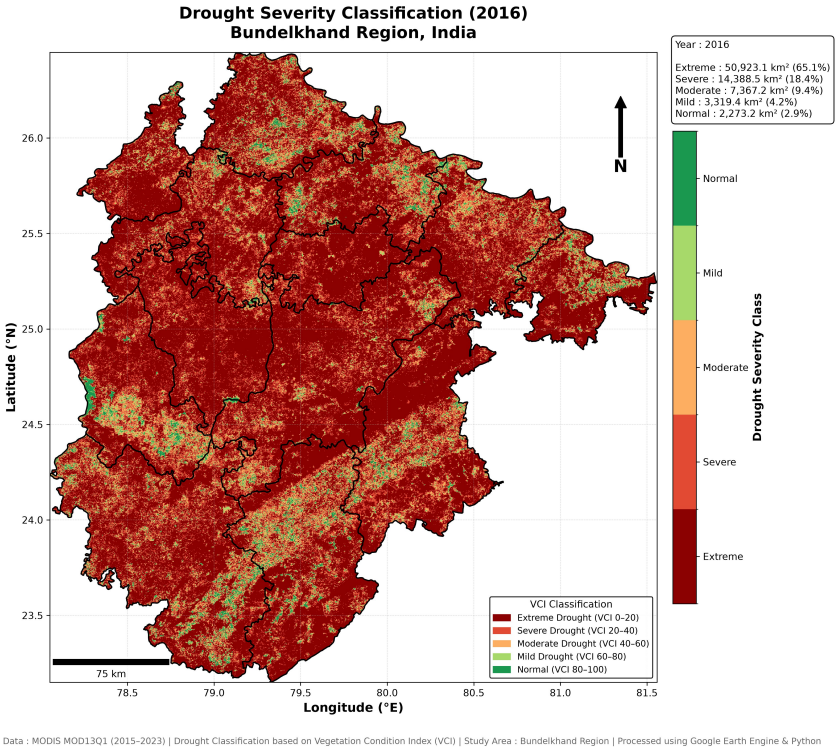

Figure 6a: Drought severity classification for 2016.

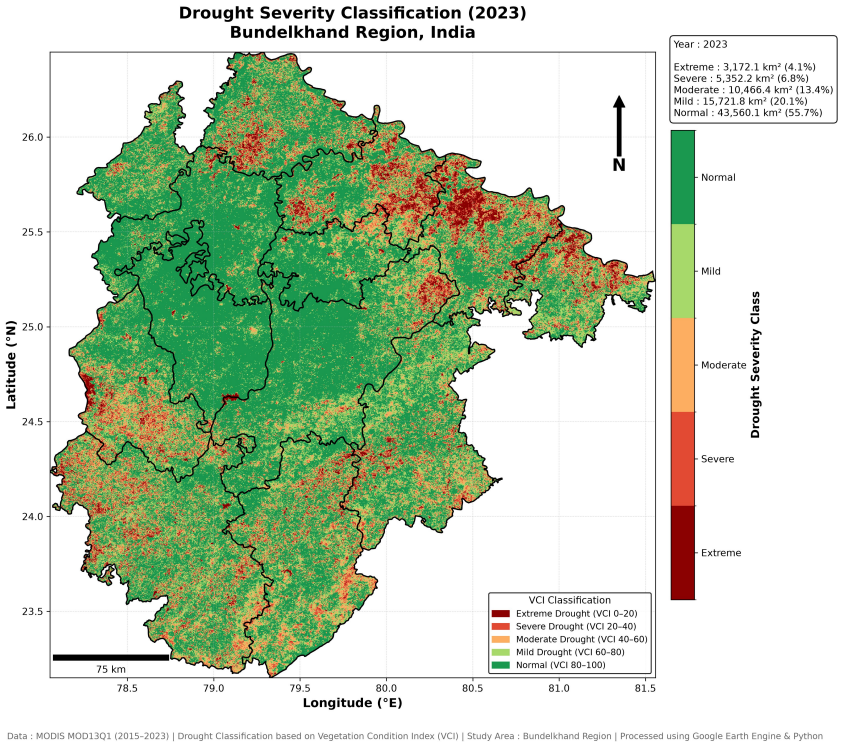

Figure 6b: Drought severity classification for 2023.

6.4 Relationship Between VCI and Rainfall

To check whether the drought pattern seen in VCI actually lines up with how much rain the region received, mean annual VCI was compared with mean annual CHIRPS rainfall using Pearson correlation, Spearman correlation and a simple linear regression (Table 3, Figure 7).

Metric	Value
Pearson correlation coefficient (r)	-0.377
Pearson p-value	0.318
Spearman rank correlation (ρ)	-0.300
Spearman p-value	0.433
Coefficient of determination (R^2)	0.142
Regression equation	VCI = -0.0376 × Rainfall + 86.39

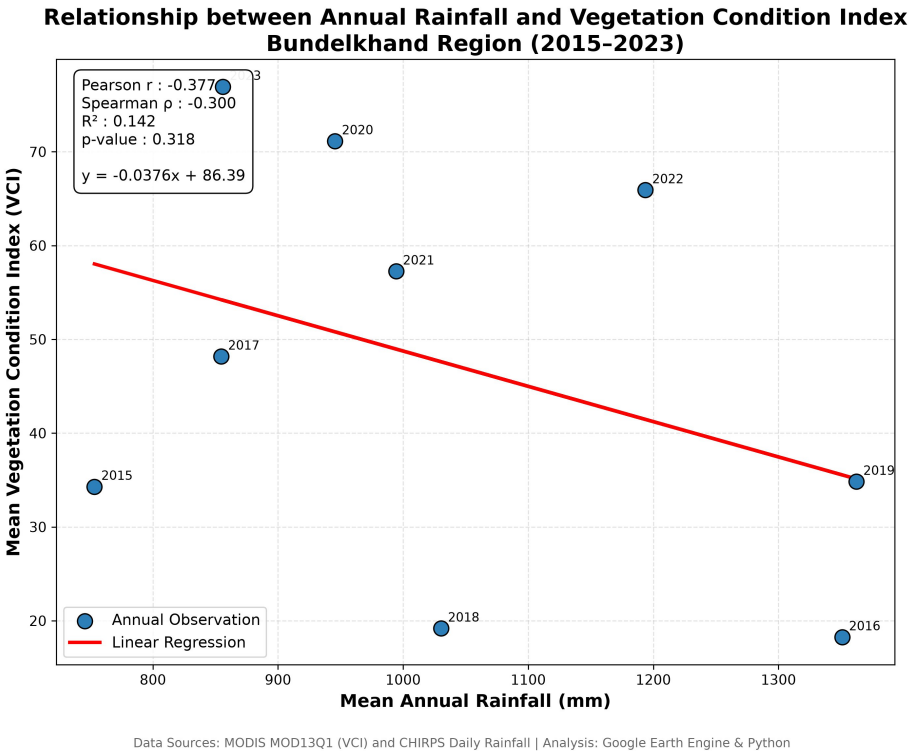

Figure 7: Scatter plot of mean annual VCI against mean annual CHIRPS rainfall (2015–2023), with regression line.

The result here was a bit surprising. Instead of a positive relationship (more rain → higher VCI, healthier vegetation), the analysis actually shows a weak negative correlation (r = -0.377), and it is not statistically significant either (p = 0.318, well above the 0.05 threshold). In simple words, based on just these nine years of data, rainfall alone does not explain the VCI pattern seen in Bundelkhand. For example, 2016 had one of the highest rainfall totals in the whole record (1351 mm) but still turned out to be the worst drought year in terms of vegetation stress. This kind of mismatch can happen because of a few reasons: rainfall might arrive in short, heavy bursts rather than being spread out evenly, so a wet annual total does not always mean the crops or natural vegetation actually benefited from it; there could be a lag between when the rain falls and when vegetation actually responds, which an annual average does not capture; and groundwater use, soil type and land management in the region probably also play a role that CHIRPS rainfall alone cannot capture. Also, with only nine data points, the correlation test does not have much statistical power, so this result should be read as indicative rather than conclusive.

7. Conclusion

This project used nine years of MODIS NDVI data (2015–2023) to work out the Vegetation Condition Index for the Bundelkhand region and to map how drought severity has changed over time. The results clearly show that 2016 and 2018 were the two worst years, with nearly half the region falling under extreme drought stress, while the region has been recovering steadily since 2020, with 2023 turning out to be the healthiest year in the entire study period.

One of the more interesting findings was that VCI did not correlate well with CHIRPS annual rainfall ($r = -0.377$, not significant). This tells us that in a region like Bundelkhand, satellite-based vegetation stress cannot be explained by rainfall totals alone, and other factors such as rainfall distribution through the year, groundwater availability, soil moisture and cropping patterns are probably influencing the outcome as well.

Overall, the VCI approach worked well as a way to track drought severity across space and time using freely available satellite data, and it can be a useful and low-cost tool for regional drought monitoring, especially in a data-scarce area like Bundelkhand. That said, the study does have a few limitations. The 250 m resolution of MODIS is coarse compared to field-level variation, so small farms or localised drought pockets may not show up clearly. The study only covers nine years, which is a fairly short record for drawing strong statistical conclusions, and no ground-truth or field survey data was used to check the VCI classification against actual crop condition on the ground.

For future work, it would help to bring in a longer NDVI time series, add soil moisture and land surface temperature data alongside rainfall, and validate the drought classes against district-level crop yield or agriculture department records. Comparing results with finer-resolution Sentinel-2 imagery for a few sample years could also help check how much detail is being lost at the 250 m MODIS scale.

8. References

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